

Concept Question 1.1 Milky Way on the night sky

From Earth we see the Milky Way as a faint band of light stretching across the night sky. Suppose the Milky Way were a band of light extending only halfway across the sky (i.e. in a semicircle). What would you conclude about the position of the Sun in the Galaxy? Explain your answer.

Problem 1.2 The Gaia survey

In this exercise, you will explore the Gaia Data Release 3 (Gaia DR3) database. You can find more details about the requested tasks in the accompanying Jupyter Notebook.

- a) Using the `astroquery.gaia` python library, create a table containing `source_id`, `ra`, `dec`, `parallax` and apparent G-band magnitude for the stars in the Gaia DR3.
- b) Plot the histograms of the apparent and absolute G-band magnitudes in your table. Why does the apparent magnitude histogram have a sharp limit?
- c) Plot the sky map of the targets in your table. Consider both equatorial and Galactic coordinates. What do the regions with the highest density of stars represent?